Step-by-Step: Deploy Streamlit + LangGraph App to OCI

1. Prepare Your Project

Your folder should look like:

```
project/
  ├── app.py
  ├── requirements.txt
  └── Dockerfile   ← (we'll create this)
```

2. Create Dockerfile (IMPORTANT)

Streamlit must run on 0.0.0.0 inside containers.

Create Dockerfile:

```
FROM python:3.10-slim

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 8501

CMD ["streamlit", "run", "app.py", "--server.port=8501", "--server.address=0.0.0.0"]
```

3. Add .dockerignore (optional but recommended)

```
__pycache__/
*.pyc
.env
.git
```

4. Build Docker Image

```
docker build -t streamlit-agent-app .
```

Test locally:

```
docker run -p 8501:8501 streamlit-agent-app
```

http://localhost:8501

5. Push to OCI Registry

Use Oracle Cloud Infrastructure Registry (OCIR).

Step 5.1: Login

```
docker login <region>.ocir.io
```

Use:

- Username: <tenancy-namespace>/<username>
- Password: Auth token (not OCI password)

Step 5.2: Tag Image

docker tag streamlit-agent-app <region>.ocir.io/<namespace>/streamlit-agent-app:latest

Step 5.3: Push Image

docker push <region>.ocir.io/<namespace>/streamlit-agent-app:latest

6. Deploy via OCI Container Instances (EASIEST)

Go to OCI Console:

👉 Developer Services → Container Instances → Create

Fill:

- **Image**: your OCIR image
- **Shape**: 1 OCPU / 2–4 GB RAM (start small)
- **Port**: 8501

7. Add Environment Variables (VERY IMPORTANT)

Your app likely needs:

- GROQ_API_KEY
- OPENAI_API_KEY (if fallback)
- Any tool configs

Add them in:
👉 Container Instance → Environment Variables

8. Open Network Access

Edit VCN security list:

- Add inbound rule:
    - Port: 8501
    - Source: 0.0.0.0/0

9. Access Your App

After deployment:

- Get Public IP from container instance
- Open:

http://<public-ip>:8501

Use OCI Generative AI Instead of Groq

You can replace Groq with:
→ OCI Generative AI

But Groq is fine if already working.